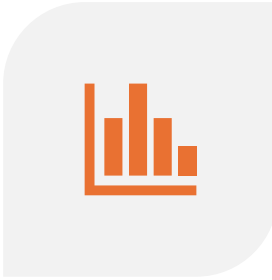
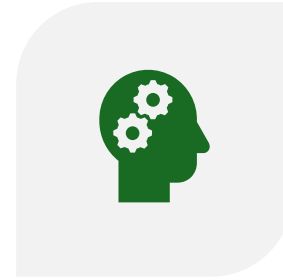


Techniques for Intrusion Detection

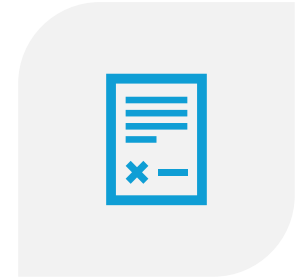
Techniques used in intrusion detection



STATISTICAL ANALYSIS



NEURAL NETWORKS AND
MACHINE LEARNING



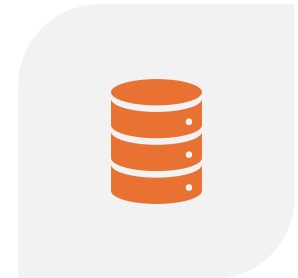
RULE-BASED AND
SIGNATURE TECHNIQUES



STATE-TRANSITION
ANALYSIS



IMMUNE SYSTEMS-
BASED TECHNIQUES



DATA MINING

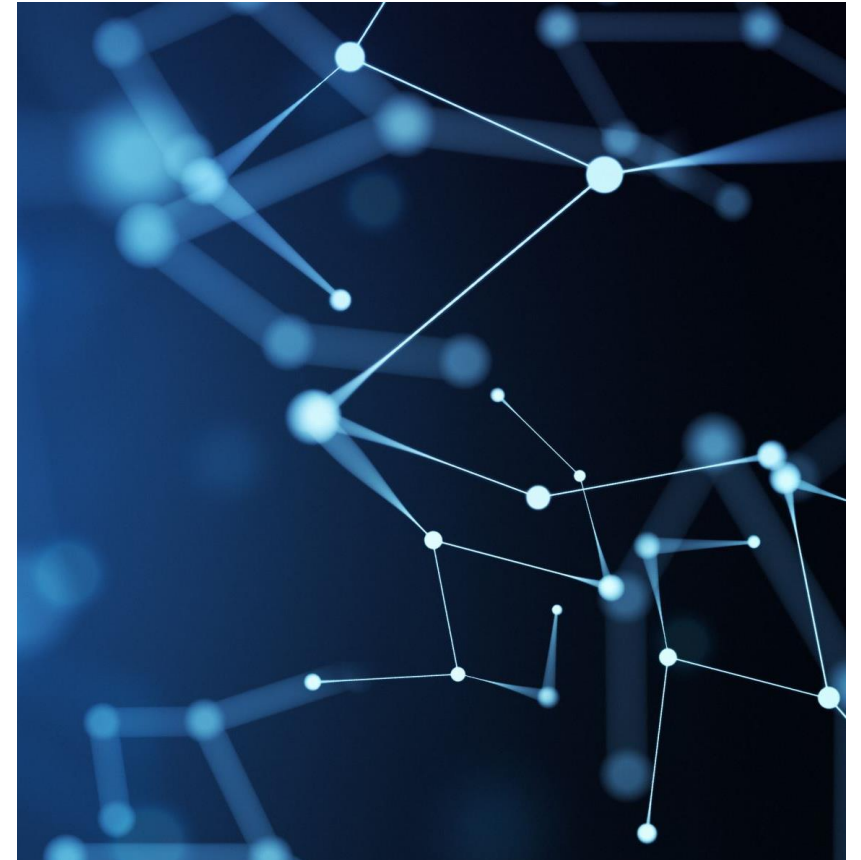


Statistical analysis approach

- The user or system behaviour (set of attributes) is measured by a number of variables over time.
- Examples of such variables are:
 - user login, logout,
 - number of files accessed in a period of time,
 - usage of disk space, memory, CPU, etc.
- The frequency of updating can vary from minutes to months
- The system stores **mean values** for each variable and detects when they exceed a predefined threshold.
- More sophisticated model: short- and long-term user profiles

Neural Networks and Machine Learning

- Neural networks use their learning algorithms to learn about the relationship between input and output.
- The main purpose is to learn the behaviour of actors in the system (e.g., users, processes).
- The advantage of using neural networks over statistics:
 - Easy way to express nonlinear relationships between variables
 - Learning about (complex) relationships automatically
- Disadvantage: NNs are still a computationally intensive technique
- It has been shown that NNs can be used to predict behaviour of users and super-users (roots)
- Other machine learning techniques may be used



Rule-based (expert) Systems



Rule-based (expert) systems: work on a previously defined set of rules describing an attack



All security related events are translated in terms of **if-then-else** rules



Inference engine maybe used to infer conclusions



Examples: Wisdom & Sense and ComputerWatch (developed at AT&T)

Signature based method

Example of a signature: `alert icmp $EXTERNAL_NET any -> $HOME_NET any (msg:"MISC large ICMP"; dsize: >800; reference:arachnids,246; classtype: bad-unknown; sid:499;)`

Alarm will be raised if a ICMP packet incoming from the external network, associated with any port and have size > 800 bytes

Advantages:

- Very low false alarm rate
- Simple algorithms, easy implementation.

Disadvantages:

- Difficulties in updating information on new types of attacks
- Unable to detect unknown attacks

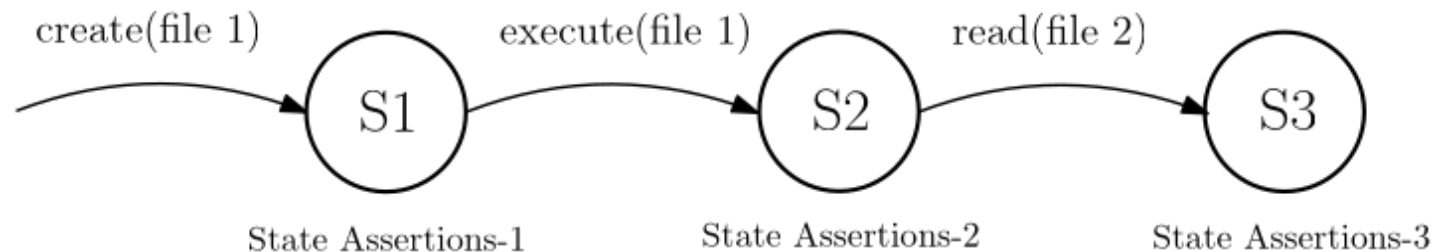
Examples: Snort, Suricata, Zeek - some of the most popular IDS

State-transition Analysis

An attack is described with a set of goals and transitions that must be achieved by an *intruder* to compromise a system.

Transitions are represented on state-transition diagrams.

Inference engine matches current audit records with attacks descriptions: event driving, forward chaining



User intention Identification



This technique models normal behaviour of users by the set of high-level tasks they have to perform on the system.



These tasks are taken as series of actions, which in turn are matched to the appropriate audit data.



If mismatch is encountered, an alarm is produced.



Computer Immunology

- Analogies with immunology → technique that constructs a model of normal behaviour of (UNIX) network services, rather than that of individual users.
- S. Forest et al., **Sense of Self for UNIX Processes**, 1996
- This model consists of *short sequences of system calls* made by the processes.
- The assumption is attacks that exploit flaws in the application code are likely to take unusual execution paths.
- The model consists of two parts:
 - audit data representing the appropriate behaviour of services
 - knowledge base with all the known “good” sequences of system calls

Computer Immunology

Stored patterns are used for monitoring of system calls to check whether the sequence generated is listed in the knowledge base; if not — an alarm is generated.

Example: pattern of normal behaviour:

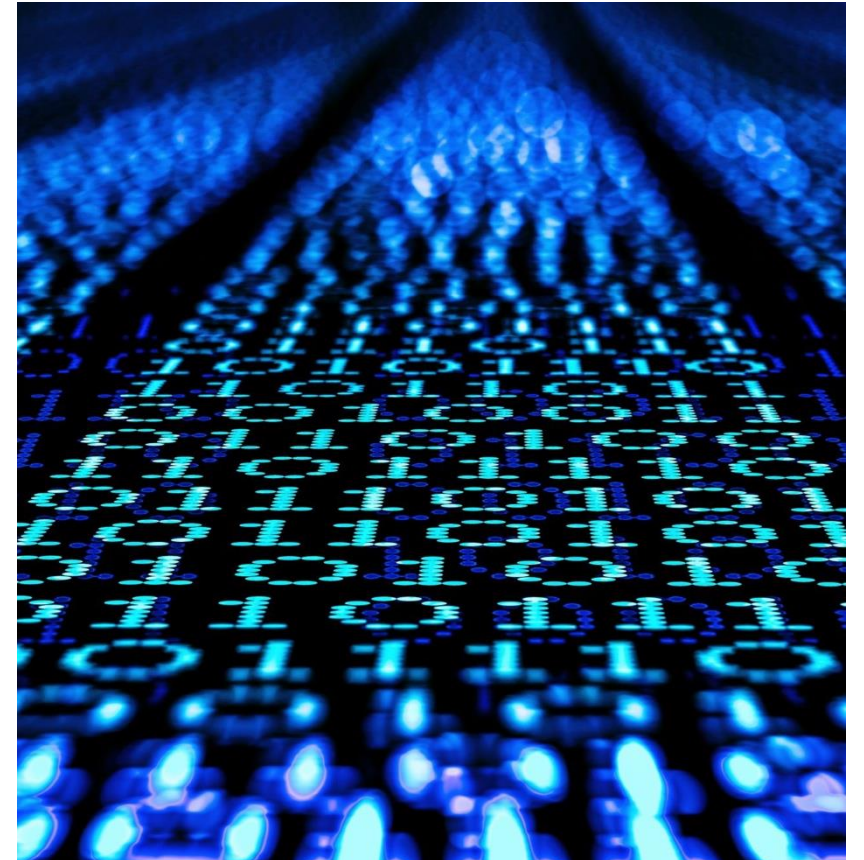
- **open, read, mmap, mmap, open, getrlimit, mmap, close** (system calls in UNIX)

Information about all sequences of the **length, say, 3** may be stored in knowledge base

If the input sequence of calls is

- **open, read, mmap, open, open, getrlimit, mmap, close**

then 4 mismatches will be found



Computer Immunology

Advantages:

- Potentially very low false alarm rate (if knowledge base complete enough)
- Updating the knowledge base can be done online

Disadvantages:

- If an attacker uses legitimate actions to gain unauthorized access, no alarm will be generated
- Arguments of system calls are not taken into account



Data mining

- Set of techniques that use the process of extracting *previously unknown* information from large stores of data.
 - Data mining techniques are suitable for large volumes of audit data (off-line processing).
 - DM is less useful for stream analysis of network traffic.
 - Based on analysis of co-occurrences and correlations of event records, DM may produce
 - Decision trees
 - Association rules
- as a concise representations of discovered relationships



Data mining

- **Example of the rule, generated by DM:**
- **if** for the past 2 seconds, the *count of connections to the same host* is greater than 4; **and** the *percentage of those that have the same service* is greater than 75%; **and** the *percentage of those that have the “S0” flag* is greater than 75%, **then** there is a *syn flood attack*.

(taken from Real Time Data Mining-based Intrusion Detection, Wenke Lee et al.)